

Sample examination papers
Generative AI and Large Language Models: mid-semester and end-semester
GenAI crash course
2026-10-10

Contents

Mid-semester paper (Units 1 to 5) **1**

End-semester paper (Units 1 to 11, weighted to Units 6 to 11) **3**

These papers are **samples** built from the question bank. The syllabus weights are: continuous assessment 30%, laboratory 20%, mid-semester 15%, end-semester 25%, capstone 10%. Numbers in the answers are computed by the course code; see the answer key.

Mid-semester paper (Units 1 to 5)

Time: 90 minutes. **Maximum marks:** 50. Answer all questions. Marks are shown after each question.

1. Name the two terms of the VAE training objective and say what each one does. *(2 marks; Unit 1)*
2. In a GAN, the discriminator is trained to: *(2 marks; Unit 1)*
 - (A) Reconstruct real images
 - (B) Tell real samples from generated ones
 - (C) Maximise the likelihood of the data
 - (D) Predict the next token
3. An autoregressive model gives the exact likelihood of a sequence but a VAE only a bound. Explain why. *(4 marks; Unit 1)*
4. Self-attention computes each output as: *(1 mark; Unit 2)*
 - (A) The average of all inputs with equal weights
 - (B) A weighted sum of value vectors with weights $\text{softmax}(\mathbf{QK}^T / \sqrt{d_k})$
 - (C) The previous output multiplied by a recurrent matrix
 - (D) The largest input
5. A Transformer block has model width $d = 768$. Using $12 d^2$ parameters per block (attention $4d^2$ plus feed-forward $8d^2$, biases ignored), how many parameters does one block have? *(2 marks; Unit 2)*

6. For sequence length $n = 2,048$ and width $d = 1,024$, give the number of floating-point operations of the attention scores and weighted sum, $4 n^2 d$. (2 marks; Unit 2)
7. Why are the scores divided by $\sqrt{d_k}$? (3 marks; Unit 2)
8. Positional encoding is needed because: (2 marks; Unit 2)
- (A) Attention is unchanged if the tokens are permuted, so order must be injected
 - (B) Embeddings are too small
 - (C) The softmax needs positive inputs
 - (D) It reduces the parameter count
9. Why can a decoder-only model generate text with a causal mask while an encoder-only model such as BERT is not used for left-to-right generation? (4 marks; Unit 2)
10. The logits are (2, 1, 0). Give the softmax probabilities at temperature $T = 1$ and at $T = 0.5$. (1 mark; Unit 3)
11. Describe the byte-pair-encoding training loop in three steps. (2 marks; Unit 3)
12. With the rule of thumb $D = 20 N$ and compute $C = 6 N D$, find the compute-optimal parameters N and tokens D for $C = 1e23$ FLOPs. (3 marks; Unit 3)
13. Using $L(N,D) = 1.69 + 406.4/N^{0.34} + 410.7/D^{0.28}$, compute the predicted loss for $N = 7e9$ and $D = 1.4e11$. (4 marks; Unit 3)
14. Five independent samples are each correct with probability 0.7, and wrong answers do not agree with each other. What is the probability that the majority answer is correct? (2 marks; Unit 4)
15. Why should the output of a model that is asked for JSON be validated, and what do you do when validation fails? (2 marks; Unit 4)
16. A chatbot keeps only the newest messages that fit in the context window. What can silently go wrong, and how do you prevent it? (4 marks; Unit 4)
17. Reciprocal rank fusion with $k = 60$ combines two rankings. A document is ranked 1st by one ranker and 3rd by the other. Give its fused score. (2 marks; Unit 5)
18. A text of 1000 characters is chunked with window 300 and overlap 50 (word boundaries respected, text of the form ‘word word ...’). How many chunks result? (2 marks; Unit 5)
19. Give three ways a RAG system can fail. (2 marks; Unit 5)
20. Recall@3 is 100% but the answers are only 90% correct. Where can the missing 10% come from, and how would you find out? (4 marks; Unit 5)

End-semester paper (Units 1 to 11, weighted to Units 6 to 11)

Time: 180 minutes. **Maximum marks:** 100. Answer all questions. Marks are shown after each question.

1. Name the two terms of the VAE training objective and say what each one does. *(2 marks; Unit 1)*
2. In a GAN, the discriminator is trained to: *(2 marks; Unit 1)*
 - (A) Reconstruct real images
 - (B) Tell real samples from generated ones
 - (C) Maximise the likelihood of the data
 - (D) Predict the next token
3. An autoregressive model gives the exact likelihood of a sequence but a VAE only a bound. Explain why. *(4 marks; Unit 1)*
4. A Transformer block has model width $d = 768$. Using $12 d^2$ parameters per block (attention $4d^2$ plus feed-forward $8d^2$, biases ignored), how many parameters does one block have? *(2 marks; Unit 2)*
5. For sequence length $n = 2,048$ and width $d = 1,024$, give the number of floating-point operations of the attention scores and weighted sum, $4 n^2 d$. *(2 marks; Unit 2)*
6. Why are the scores divided by $\sqrt{d_k}$? *(3 marks; Unit 2)*
7. Why can a decoder-only model generate text with a causal mask while an encoder-only model such as BERT is not used for left-to-right generation? *(4 marks; Unit 2)*
8. The logits are $(2, 1, 0)$. Give the softmax probabilities at temperature $T = 1$ and at $T = 0.5$. *(1 mark; Unit 3)*
9. With the rule of thumb $D = 20 N$ and compute $C = 6 N D$, find the compute-optimal parameters N and tokens D for $C = 1e23$ FLOPs. *(3 marks; Unit 3)*
10. Five independent samples are each correct with probability 0.7, and wrong answers do not agree with each other. What is the probability that the majority answer is correct? *(2 marks; Unit 4)*
11. A chatbot keeps only the newest messages that fit in the context window. What can silently go wrong, and how do you prevent it? *(4 marks; Unit 4)*
12. Reciprocal rank fusion with $k = 60$ combines two rankings. A document is ranked 1st by one ranker and 3rd by the other. Give its fused score. *(2 marks; Unit 5)*
13. A text of 1000 characters is chunked with window 300 and overlap 50 (word boundaries respected, text of the form ‘word word ...’). How many chunks result? *(2 marks; Unit 5)*
14. Recall@3 is 100% but the answers are only 90% correct. Where can the missing 10% come from, and how would you find out? *(4 marks; Unit 5)*
15. QLoRA differs from LoRA in that the frozen base weights are stored: *(1 mark; Unit 6)*
 - (A) In 4-bit precision
 - (B) As one-hot vectors
 - (C) In a vector database
 - (D) Only on the CPU

16. A weight matrix is 4096 x 4096. How many trainable parameters does a rank-8 LoRA adapter add, and what fraction of the matrix is that? (2 marks; Unit 6)
17. How much memory do the weights of a 7-billion-parameter model need at 16 bits and at 4 bits (weights only)? (2 marks; Unit 6)
18. What is catastrophic forgetting and name one way to reduce it. (3 marks; Unit 6)
19. In supervised fine-tuning the loss is averaged over the response tokens only. Why mask the prompt tokens? (4 marks; Unit 6)
20. A vision transformer cuts a 224 x 224 image into 16 x 16 patches. How many image tokens? How many for a 336 x 336 image with 14 x 14 patches? (1 mark; Unit 7)
21. CLIP-style training makes matching image-text pairs: (2 marks; Unit 7)
- (A) Close in a shared embedding space and non-matching pairs far apart
 - (B) Identical pixels
 - (C) Share the same tokenizer
 - (D) Have the same length
22. State the closed form of the forward diffusion step and why it is useful for training. (3 marks; Unit 7)
23. What does the guidance weight w do in classifier-free guidance, and what is the cost of a large w ? (4 marks; Unit 7)
24. In a ReAct-style agent loop the model alternates: (1 mark; Unit 8)
- (A) Training and evaluation
 - (B) Reasoning, choosing an action (tool call), and reading the observation
 - (C) Encoding and decoding
 - (D) Retrieval and chunking
25. An agent must complete a chain of 10 steps. Each step succeeds with probability 0.95, independently. What is the probability that the whole chain succeeds? What if each step succeeds with 0.99? (2 marks; Unit 8)
26. Give two defences against prompt injection in a tool result. (3 marks; Unit 8)
27. Five agents exchange messages for one round. In a fully connected design everyone tells everyone; in a hub design a manager talks to each worker and each worker replies. Count the messages in each design. (4 marks; Unit 8)
28. BLEU mainly measures: (1 mark; Unit 9)
- (A) Factual correctness
 - (B) Overlap of n-grams between a candidate and a reference
 - (C) Toxicity
 - (D) Latency
29. Two raters label 10 answers as good (1) or bad (0): $A = [1, 1, 1, 0, 0, 1, 0, 1, 1, 0]$, $B = [1, 1, 0, 0, 0, 1, 1, 1, 1, 0]$. Compute Cohen's kappa. (2 marks; Unit 9)

- 30.** A system answers 18 of 20 questions correctly. Give the 95% Wilson interval for its accuracy. *(2 marks; Unit 9)*
- 31.** Name a bias of an LLM judge and a simple way to reduce it. *(3 marks; Unit 9)*
- 32.** The Laplace mechanism adds noise with scale $b = \text{sensitivity} / \epsilon$. For sensitivity 1 and $\epsilon = 0.5$, give b and the standard deviation of the noise. *(4 marks; Unit 9)*
- 33.** Continuous batching improves serving because: *(1 mark; Unit 10)*
- (A) Finished requests leave the batch and new ones join at once
 - (B) It makes each request faster
 - (C) It reduces the model size
 - (D) It removes the KV cache
- 34.** A model has 32 layers, 8 key-value heads of width 128 (hidden 4096, 32 heads) and a 16-bit cache. Bytes of KV cache per token, and for a 2000-token sequence? *(2 marks; Unit 10)*
- 35.** A rented GPU costs 2 dollars per hour, generates 1000 tokens per second when busy, and is busy 50% of the time. Dollars per million tokens? *(3 marks; Unit 10)*
- 36.** Why report the 95th-percentile latency instead of the mean? *(2 marks; Unit 10)*
- 37.** List the five parts of a design brief and say why it is written before the code. *(2 marks; Unit 11)*
- 38.** 34 of 40 answers are correct. Give the 95% Wilson interval. *(2 marks; Unit 11)*
- 39.** A legal assistant judges a wrong answer 9 times as costly as a refusal. Above what probability that the documents contain the answer should it answer? *(3 marks; Unit 11)*
- 40.** You chose a refusal threshold on your evaluation questions and report the result on the same questions. What is wrong, and what do you do instead? *(4 marks; Unit 11)*